Evaluating Methods to Identify Introgressed Regions

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BACKGROUND

Introduction

STRUCTURE[1] is a program used to identify population composition based on provided data. However, certain population histories cause these programs to incorrectly infer population formation. Computational biologists created *ms*[2] and *msprime*[3] to simulate population histories and test the performance of STRUCTURE.

Because STRUCTURE is used with real population data, it is important to work with an accurate program to correctly infer coalescent events. Knowing the effects will allow biologists to know when to approach these population histories with caution.

Objective

This study focuses on simulating population histories that will make STRUCTURE produce statistically different admixture proportions.

Hypotheses

- Adjusting admixture time affects STRUCTURE output
- Ghost admixture will affect the proportions output by STRUCTURE
- Adjusting the ghost admixture time will affect STRUCTURE output

METHODS

Simulate population data using *msprime*

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Create data files containing SNP locations and genotypes using python script

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Run STRUCTURE analysis using the data files created

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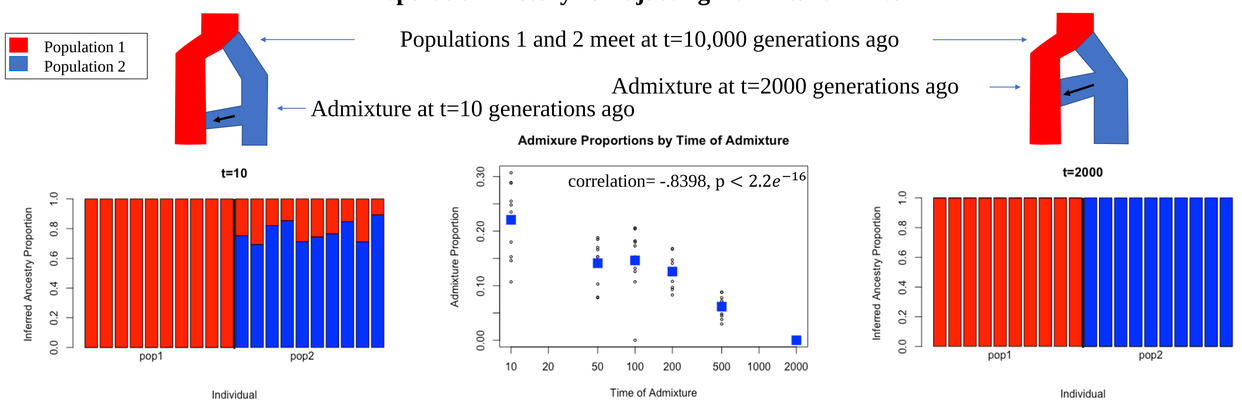
Take STRUCTURE output and create plots using R

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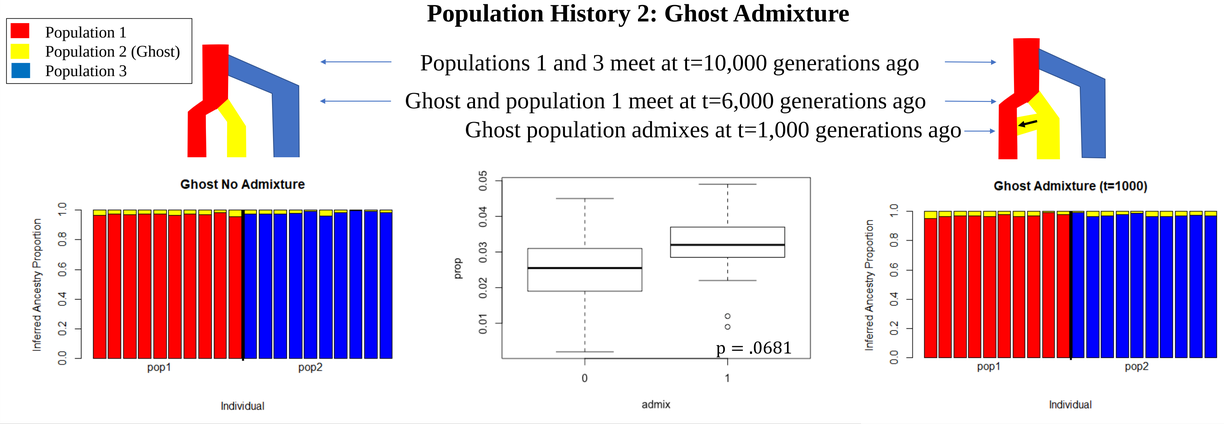
Conduct statistical analyses (t-test, correlation test, and ANOVA)

RESULTS

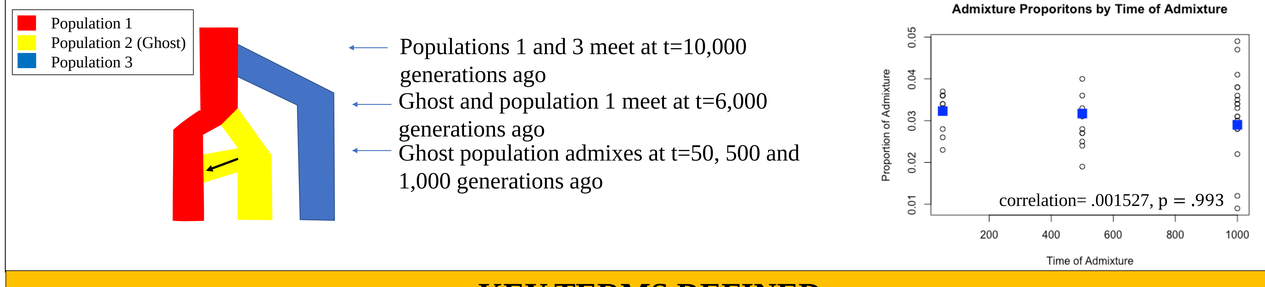
Population History 1: Adjusting Admixture Times



Population History 2: Ghost Admixture



Population History 3: Adjusting the Ghost Admixture Time



KEY TERMS DEFINED

Introgression – The introduction of genes from one species into the gene pool of another species. This occurs when mating between two individuals produces fertile hybrids.

Admixture – Two or more previously isolated populations interbreeding.

Ghost Population – An unsampled population.

Bottlenecking – An event when a population suddenly decreases in size, and after a short amount of time, the same population suddenly increases in size.

INTERPRETATION

History 1: A strong negative correlation between the admixture time and proportions output by STRUCTURE may result from the fact that more time allows for individuals to mix genetic information with each other. This may cause STRUCTURE to give lower proportions than expected as time increases.

History 2: Because STRUCTURE observed mixture in both the admixed population and third population when it was not admixed with the ghost, it appears that STRUCTURE cannot easily identify ghost populations. Perhaps the model is not well specified.

History 3: A lack of correlation between ghost admixture times and proportions from STRUCTURE could be due to the limited number of times simulated. In our test, only t=50, 500 and 1,000 generations were tested, so more times should be simulated.

IMPLICATIONS

Understanding the time of admixture affects the proportion STRCTURE outputs allows computational biologists to know we need to approach recent admixture scenarios with caution.

Since ghost admixture does not have a significant effect on STRUCTURE output, we know we can rely on STRUCTURE to give consistent results.

FUTURE RESEARCH

To understand ghost admixture, we would like to explore the effects of a bottlenecking scenario with a ghost population since groups of people that go through bottlenecking are more genetically similar.

We would also like to understand how phasing data through Beagle[4] can affect STRUCTURE output.

ACKNOWLEDGEMENTS

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